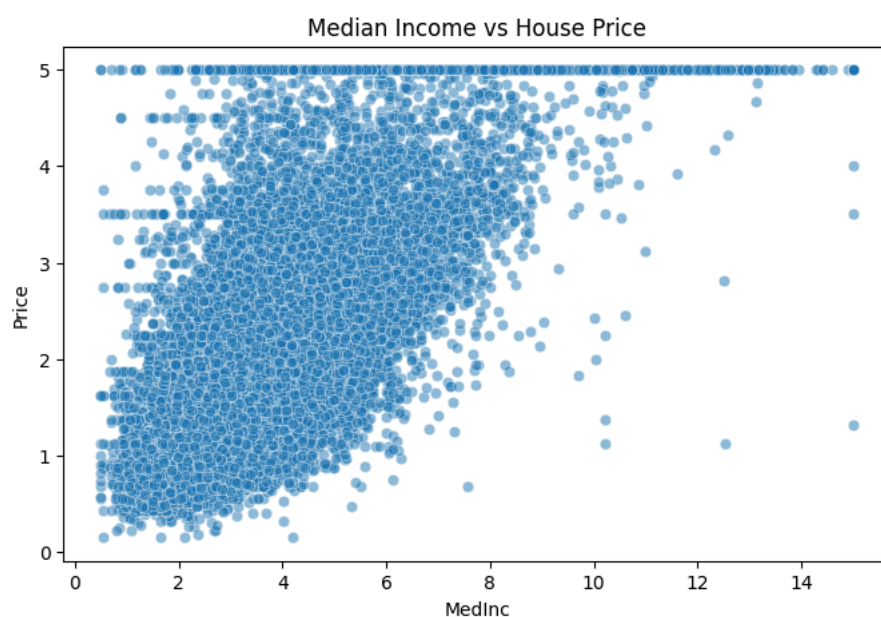


```
# 1.) Importing the necessary libraries.
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
# 2. Loading the dataset.
california = fetch_california_housing()
df = pd.DataFrame(california.data, columns=california.feature_names)
df['Price'] = california.target
```

```
# 3.) Exploratory Data Analysis (EDA).
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='MedInc', y='Price', alpha=0.5)
plt.title('Median Income vs House Price')
plt.show()
```



```
# 4.) Splitting the data.
X = df.drop('Price', axis=1)
y = df['Price']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
# 5.) Training the Regression Model.
model = LinearRegression()
model.fit(X_train, y_train)
```

LinearRegression ⓘ ?

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LinearRegression()
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```
# 6.) Making the predictions.
y_pred = model.predict(X_test)
```

```
# 7.) Evaluation.
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"Mean Absolute Error: {mae:.4f}")
print(f"R-squared Score: {r2:.4f}")
```

```
Mean Absolute Error: 0.5332
R-squared Score: 0.5758
```

```
# 8.) Visualization.
plt.figure(figsize=(8, 5))
```

```
plt.scatter(y_test, y_pred, alpha=0.5, color='green')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw=2)
plt.xlabel('Actual Price')
plt.ylabel('Predicted Price')
plt.title('Actual vs Predicted House Prices')
plt.show()
```

